Answers

1.**A**	2.**D**	3.**B**	4.**D**	5.**A**	6.**D**	7.**C**	8.**A**

Explanations

1.**A**

When total matchstick is 1, B will surely win as A is the one who is picking up matchstick first.

When total matchstick is 2, A will surely win as A will pick one matchstick then B has to pick last one.

When total matchstick is 3, A will surely win as he takes up a strategy so that B will have last matchstick to pick up. So A will pick 2 matchstick.

When total matchstick is 4, B will surely win as if A picks up one matchstick, remaining matchsticks will be 3 and we know that when total 3 matchsticks are there then the one, who is picking up matchstick first, is winning surely.

When total matchstick is 5, A will surely win as he takes up a strategy so that B will have last matchstick to pick up. So A will pick 1 matchstick first, then 4 matchsticks will be there and we know that the one, who picks up matchstick second now, will win. Hence A will win.

When total matchstick is 6, A will surely win either he takes up a strategy so that B will have last matchstick to pick up. So A will pick 2 matchstick first, then 4 matchsticks will be there and we know that the one, who picks up matchstick second now, will win. Hence A will win.

Hence it is making a pattern when number of matchsticks are 1,4,7,10......N then B is winning the game.

So here minimum value of N greater than 5 will be 7.

VIEW SOLUTION

2.**D**

When total matchstick is 1, B will surely win as A is the one who is picking up matchstick first.

When total matchstick is 2, A will surely win as A will pick one matchstick then B has to pick last one.

When total matchstick is 3, A will surely win as he takes up a strategy so that B will have last matchstick to pick up. So A will pick 2 matchstick.

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Hence it is making a pattern when number of matchsticks are 1,4,7,10......N then B is winning the game.

So largest value of N (less than 50) where B will win, is 49.

VIEW SOLUTION

3.**B**

From the given passage, it is given that Q>P, R>S. Now it is given that R is greater than or equal to S+2. Option B will still remain true because it is given that Q>P.

VIEW SOLUTION

4. **D**
From the given passage, it is given that Q>P, R>S.

According to the information Q = 2 when P = 1 or Q = 4 when P = 2.

In first case R = 4 and S = 3 and in the second case R = 3 and S = 1.

In both the instances S is odd.

VIEW SOLUTION

5. **A**
From the given passage, it is given that Q>P, R>S.

When R = 4 and S = 3, then Q = 2 and P = 1.

P+Q = 2+1 = 3 which contradicts statement 1.

VIEW SOLUTION

6. **D**
Number of coins with A = 54

Number of coins with B = 2x+2

Number of coins with C = x

Number of coins with D = y

54+3x+2+y = 100

3x+y = 44

If 2x+2 = 28, then x = 13 and y = 5 which is not possible.

If 2x+2 = 20, then x = 9 which is not possible.

If 2x+2 = 26, then x = 12 and y = 8 which is not possible

If 2x+2 = 22, then x = 10 and y = 14 which is feasible

VIEW SOLUTION

7. **C**
If A collected 54 coins, remaining will be 46.
Now B,C,D must have atleast 10,12,14 coins
So remaining coins will be 10
So difference between maximum and second maximum should be least when second highest will be as maximum as possible.
I.e. difference = 54- (14+10) = 30

VIEW SOLUTION

8. **A**
As least number of coin, one can have is 10
And no two of them possess same number of coins, hence for having maximum number of coins to one person, the distribution of coins will be = 10,12,14,64
maximum no. of coins = 64

VIEW SOLUTION